

Linear Equation Art Studio

A creative project where math and design come together. Use slope-intercept form to draw original artwork — every line you place is a real linear equation.

$$y = mx + b$$

SLOPE
Y-INTERCEPT

Quick Review: Slope-Intercept Form

Every non-vertical line in the coordinate plane can be written as $y = mx + b$, where:

m

SLOPE — how steep the line is and which direction it tilts.
Calculate as **rise ÷ run** (change in y over change in x).
Positive m: line tilts up-right. **Negative m**: line tilts down-right. **m = 0**: horizontal.

b

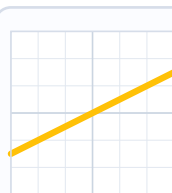
Y-INTERCEPT — the y-coordinate where the line crosses the y-axis.
It's the value of y when x = 0. On a graph, it's the line's vertical "starting point" at the y-axis.

Slope at a Glance



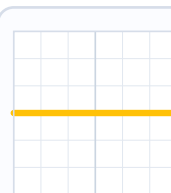
Steep, Positive

$m = 2$: rise 2 for every 1 right.



Gentle, Positive

$m = 1/2$: rise 1 for every 2 right.



Zero (Horizontal)

$m = 0$: y stays the same as x changes.



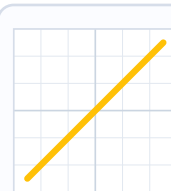
Gentle, Negative

$m = -1/2$: down 1 for every 2 right.



Steep, Negative

$m = -2$: down 2 for every 1 right.



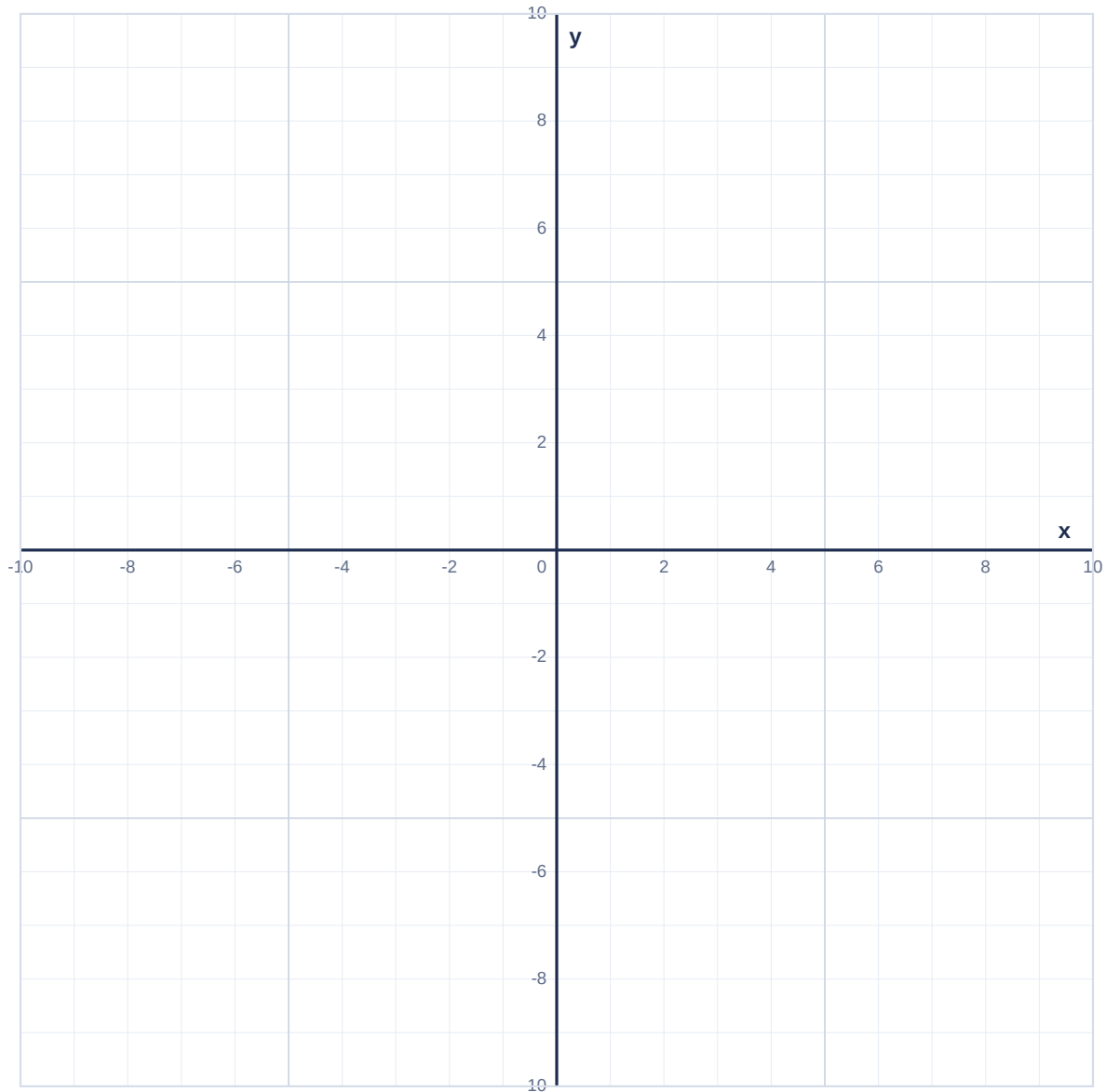
Diagonal $m = 1$

Classic 45-degree line. Up 1, right 1.

Step 1 · Plan Your Artwork

Sketch your design here BEFORE you start using the studio. Each line you draw will become an equation in slope-intercept form. Use a pencil so you can adjust as ideas develop.

Tips: Mark where each line crosses the y -axis (that gives you b). Estimate the slope by counting boxes — rise over run.



Step 2 · Record Your Equations

As you add each line in the studio, record its equation here. Identifying the slope's sign and steepness builds fluency with linear functions.

#	Equation ($y = mx + b$)	Slope (m)	y-int (b)	Sign of m $+ / - / 0$	What It Represents
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					

Bonus check: Are any of your lines parallel? Lines are parallel when they have the same slope (m) but different y-intercepts (b). Are any perpendicular? Perpendicular lines have slopes that multiply to -1 .

Step 3 · Reflect on the Math

Answer each question in 1–3 complete sentences. Use vocabulary from class: *slope*, *y-intercept*, *parallel*, *perpendicular*, *positive*, *negative*, *steep*, *gentle*, *horizontal*.

1. Look at the equations you used. Which line has the STEEPEST slope? How do you know just by looking at the m-value (without graphing)?

2. Did you use any parallel lines (same slope, different y-intercepts)? If yes, list two of their equations. If no, what would you change to add a pair?

3. Pick one line whose y-intercept (b) is NOT zero. Where does that line cross the y-axis? Connect b to what you SEE on the canvas.

4. What's the difference between a line with slope $\frac{1}{4}$ and one with slope 4? Describe how each one looks on the canvas.

5. Imagine doubling the slope of your favorite line. What would happen visually? Explain whether the y-intercept changes.

6. If you could only use lines with negative slopes, what kinds of art would still be possible? What would NOT be possible?

Bonus · Challenge Cards

Pick one or more challenges to push your skills. Each one targets a specific concept about slope or y-intercept.

Mountain Range

Use 5 lines with mixed positive and negative slopes to make a mountain skyline.

■ **Concept: comparing positive vs. negative slopes.**

Starburst

Place 8 lines that all pass through the origin ($b = 0$). Use a wide range of slopes — both signs.

■ **Concept: when $b = 0$, the line passes through $(0, 0)$.**

Sunset Scene

Place a horizon ($m = 0$), then add 5 "sun rays" radiating from a single high point. Use warm colors.

■ **Concept: lines through a common point with varied slopes.**

Perpendicular Pair

Add two lines whose slopes multiply to -1 (for example, $m = 2$ and $m = -1/2$). They cross at right angles.

■ **Concept: perpendicular lines.**

Rainbow Stairs

Draw 4 parallel lines (same m , different b) using 4 different colors stacked above each other.

■ **Concept: parallel lines have equal slopes.**

Block Letter

Spell the first letter of your name using ONLY horizontal lines ($m = 0$) and very steep lines ($|m| \geq 6$).

■ **Concept: horizontal vs. nearly-vertical extremes of slope.**

Gentle vs Steep

Draw one line with $|m|$ less than 1 and one with $|m|$ greater than 3. Pick contrasting colors.

■ **Concept: comparing slope magnitudes.**

Line Family

Draw 4 lines that all have the same y-intercept (b) but different slopes. They'll all cross at one point.

■ **Concept: same b , different $m \rightarrow$ all share the y-intercept point.**

Final Showcase Checklist

- I planned my artwork on graph paper before opening the studio.
- I used at least 8 different equations.
- I included at least one positive, one negative, and one zero slope.
- I recorded every equation I used in the tracking table.
- I answered all six reflection questions in complete sentences.
- I tried at least one challenge from this page.
- I exported my final art as a PNG and saved it.